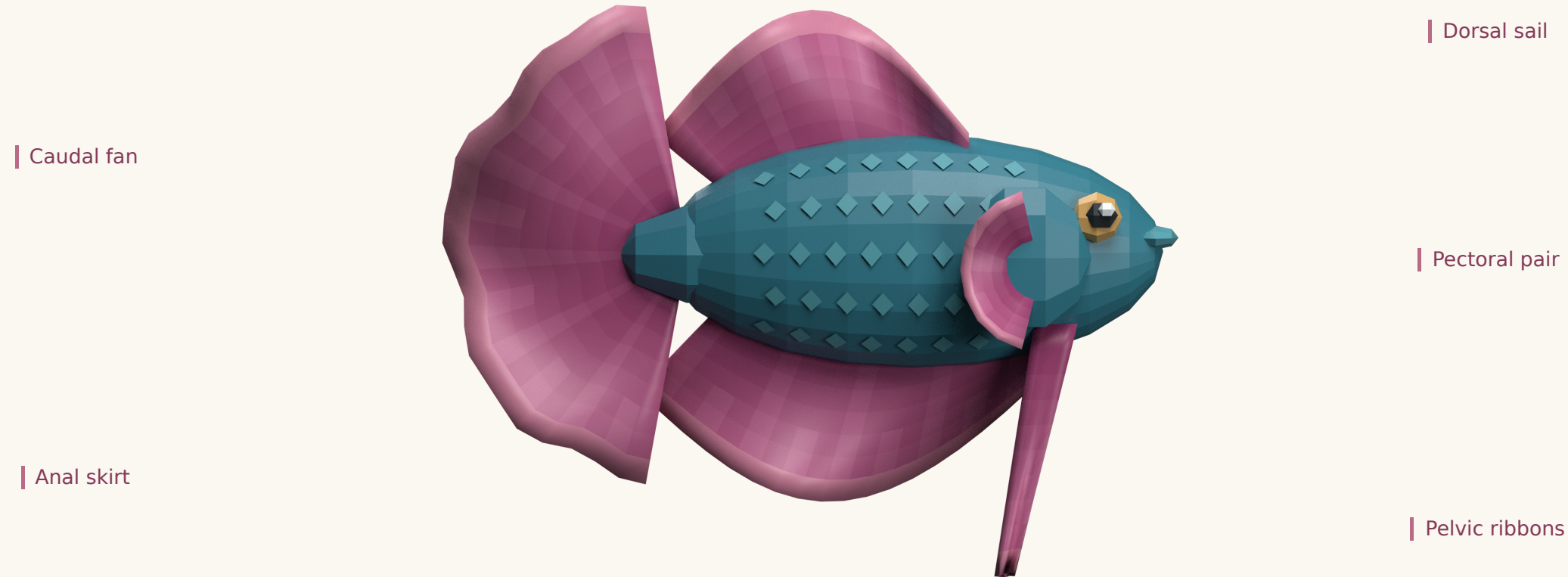


Betta — the art of flowing fins

Betta splendens · Ornamental long-fin form · Family Osphronemidae



Runtime fish • 4,076 triangles • eight mesh groups • opaque membranes.
Same geometry and fin deformation as the app; smooth studio lighting.

Centuries of breeding [1]

Historical reports reach the 14th century; an exact domestication date is uncertain.

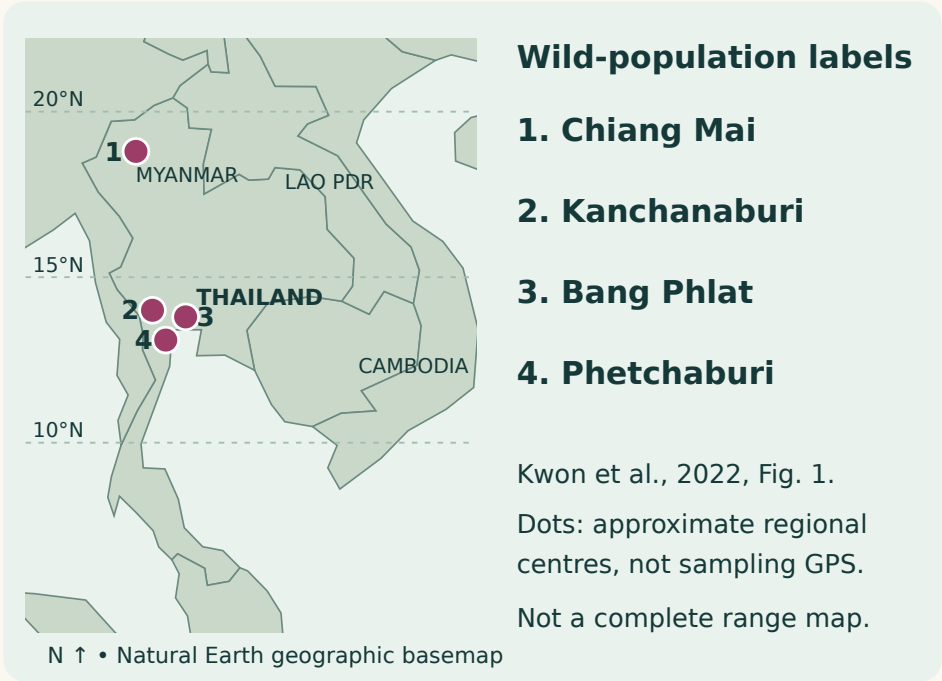
Early 20th century [1]

Ornamental breeding expands the range of colours and fin shapes.

5 February 2019 [2]

Declared Thailand's National Aquatic Animal.

Wild betta locations [1]



SHALLOW FRESHWATER

Vegetation and calm water

Floodplain pools, canals, rice paddies, swamps and slow streams are among its recorded habitats. Wild fish have compact fins; the large fins above belong to an ornamental form. [1, 2]

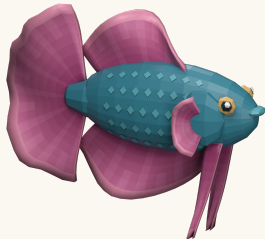
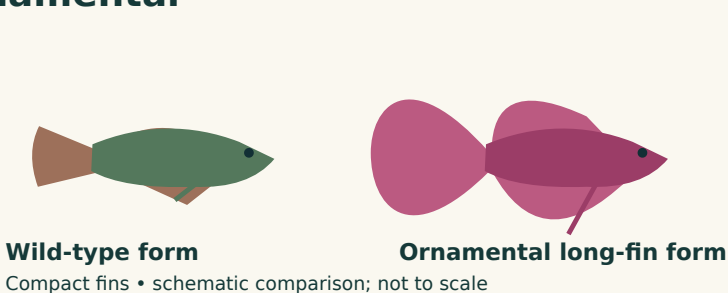
Air at the surface

A labyrinth organ allows air breathing alongside the gills, helping in low-oxygen water. Surface access matters. [2]

A bubble-nest father

Male *B. splendens* build bubble nests and care for eggs and newly hatched fry. Other *Betta* species may mouthbrood instead. [3]

Wild and ornamental



Same mesh, oblique view.
Seven fins, including both paired sets.

How the fins flow

GAME STUDY • flexible surfaces, anchored roots

HOVER



Pectoral scull

SWIM



Tips trail drive

TURN



Asymmetric fins

SETTLE



Damped recovery

Keep fin roots attached; allow curved rays, trailing edges and delayed tips.

Ray flexibility is supported by general fish research in **bluegill**. The sequence illustrates game motion, not measured betta kinematics. [4]

Sources: [1] Kwon et al., Science Advances 8:eabm4950 (2022), Fig. 1 · [2] Sermwatanakul, SEAFDEC (2019) · [3] Panijpan et al., Thai Natural History Museum Journal 11(1) (2017) · [4] Flammang et al., J. Morphology (2013).

Original fish geometry, renders and diagrams. Natural Earth basemap: public domain. Regional dots are not collecting coordinates or a complete/current range. The ornamental model does not represent the wild phenotype at those sites.

A3 portrait · 297 × 420 mm · Research checked 2 October 2026 · Editable model and source notes accompany this poster.